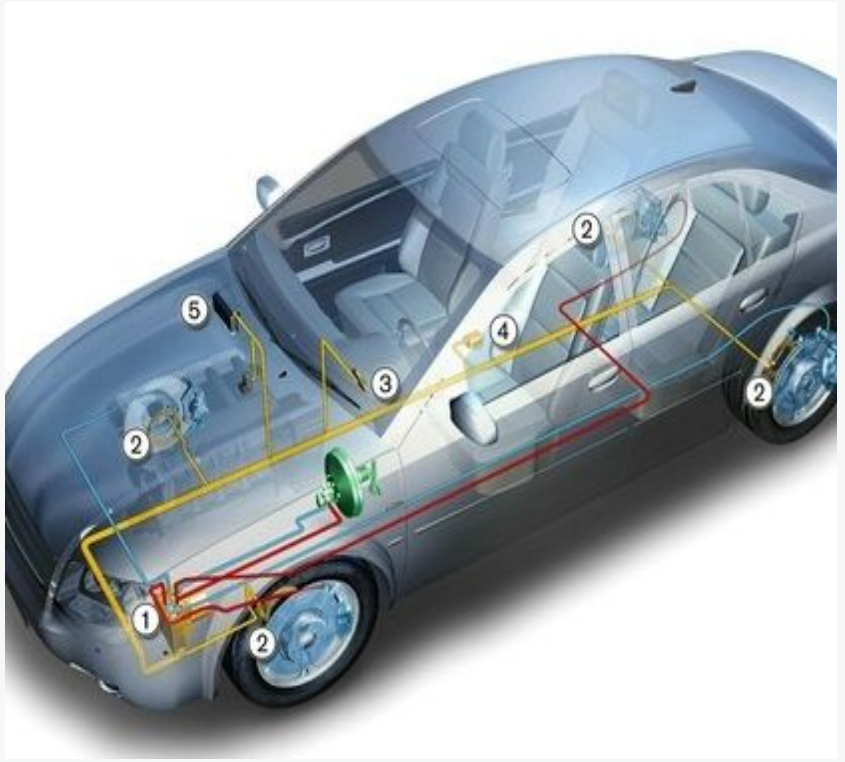
Securing In-Vehicle Networks: A Post-Quantum and Federated Learning Architecture



Overview

This project proposes a segmented CAN-FD architecture that integrates Post-Quantum Cryptography and machine learning.

It utilizes hardware-rooted identity and ML-DSA for remote ECU attestation.

The system transfers massive post-quantum payloads using a custom 60/4 indexed block-transfer mechanism over CAN-FD.

It implements a Federated Learning-based Intrusion Detection System at the Gateway to detect anomalies locally.

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Research Motivation

With the age of autonomous fleets, we will see more and more vehicles which will have little to no human interaction and control. This also poses a greater security risk as all the vehicles ina fleet are connected in one way or another.

The standard Controller Area Network protocol lacks encryption and authentication, leaving vehicles exposed to remote attacks.

The growing threat of quantum computing breaks traditional vehicle cryptography, demanding a shift to post-quantum resilience for long-lifespan vehicles.

Legacy CAN protocols fail under the massive payload sizes of explicit post-quantum certificates.

Cryptography alone cannot prevent network flooding or spoofing, requiring a dynamic machine learning detection layer.

Clearly, there is a greater and immediate need for proper study in this area for new deployable technologies.

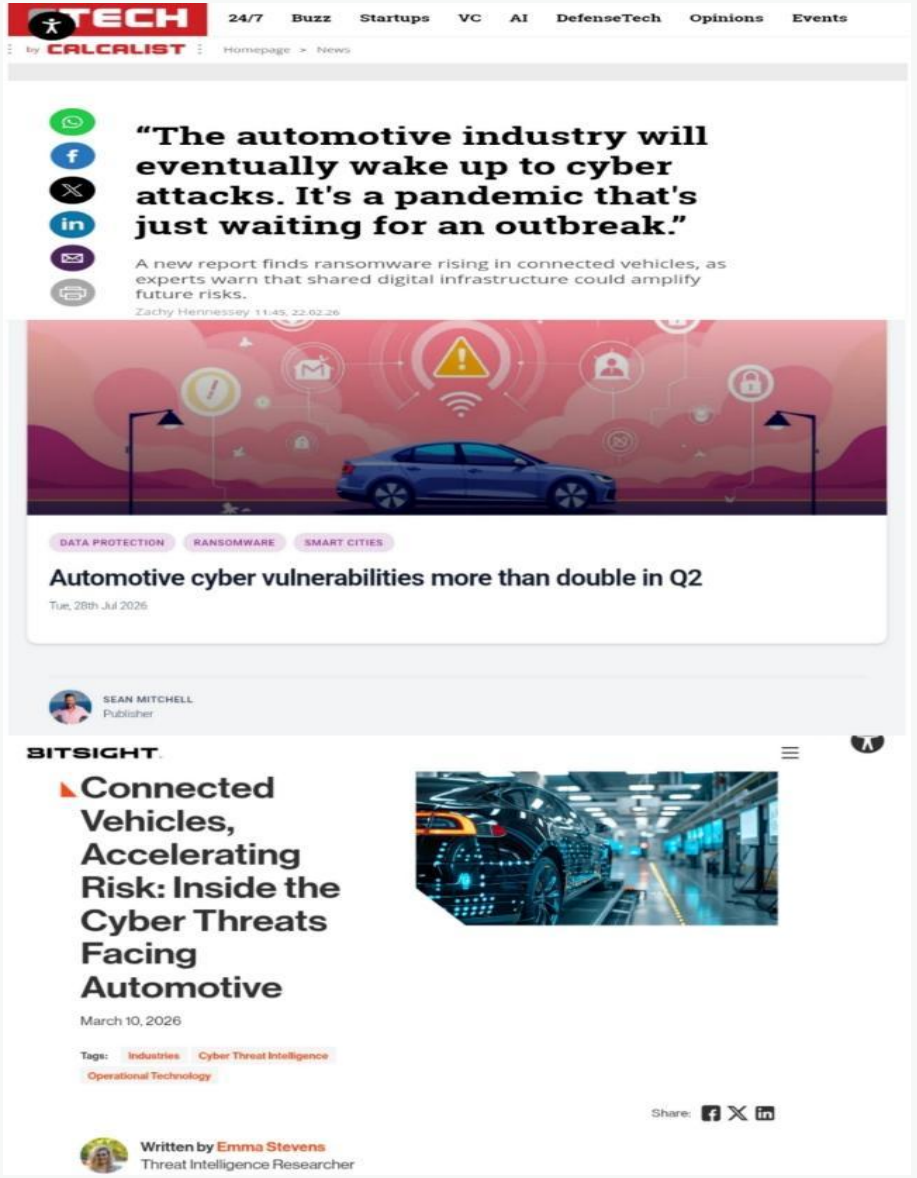
Evidence and Industry Context

High-profile attacks, such as the 2015 remote hijacking of a Jeep Cherokee's braking and steering, prove that in-vehicle network vulnerabilities are a critical safety risk.

The architecture addresses the rigorous requirements of the ISO/SAE 21434 standard ("Road vehicles — Cybersecurity engineering") by enforcing data confidentiality and mitigating emerging threats.

The integration of ML-DSA directly aligns with the recent NIST FIPS 204 standard for post-quantum digital signatures.

Current research highlights the lack of post-quantum payload handling in standard protocols; our simulated CAN-FD block-transfer mechanism directly solves this documented gap.



Proposed Methodology

Step 1: Network Architecture and Cryptography

The architecture divides the network into functional domains connected by a Central Gateway and establishes device trust using reconstructed device-root secrets.

The Gateway initiates startup with a fresh challenge, and the ECU responds with a post-quantum attestation signature.

The system slices large attestation objects into fragments, transmitting them with ordering and status indicators across CAN-FD frames.

The framework wraps all runtime telemetry using authenticated encryption, applying replay protection and sender identity validation.

Step 2: Gateway-Side Intrusion Detection

The system prepares structured time-series features from decrypted telemetry and message behavior.

A lightweight anomaly-detection model runs locally on the Gateway to flag malicious traffic without relying on encrypted payloads.

Federated Learning aggregates model updates across a simulated multi-vehicle fleet, keeping raw telemetry strictly local.

Step 3: Integrated Evaluation

The project validates the complete software lifecycle from authenticated startup to local intrusion detection inference.

It profiles system performance by measuring CPU, memory, and latency overhead to provide concrete comparative benchmarks for publication.

Post-Quantum Cryptography for ECU Security Use Cases

WHAT THE PAPER ENABLES

- Shows PQC signatures are practical on constrained automotive microcontrollers.
- Identifies Dilithium as a strong performance/security candidate for secure updates.

WHERE IT STOPS

- XMSS requires strict state management and limits signatures per key.
- No AES hardware acceleration; XMSS hashing was slower than optimized implementations.

RESEARCH GAP

Validate PQC under real-time CAN constraints and assess integration with EVITA-compliant hardware security modules.

CAPSTONE LINK

Benchmark quantum-safe authentication on real or hardware-in-the-loop automotive gateways.

PQ-CAN: Simulating Post-Quantum Cryptography in Embedded Systems

WHAT THE PAPER ENABLES

- Scalable, modular test environment with adjustable CPU and CAN bit-rate constraints.
- Makes security-strength, latency, and delivery-success trade-offs explicit.

WHERE IT STOPS

- Software-only simulation; no validation on physical CAN hardware or legacy ECUs.
- Large keys/signatures caused latency and timeouts under heavy traffic.

RESEARCH GAP

Extend PQ-CAN to physical networks and optimize algorithms that miss time-sensitive automotive deadlines.

CAPSTONE LINK

Use simulation for selection, then verify chosen primitives on a gateway testbed.

Quantum Federated Learning for Autonomous Vehicle Cybersecurity

WHAT THE PAPER ENABLES

- Addresses non-IID learning, privacy, Byzantine clients, and quantum-era interception together.
- Quantum distance estimation promises fleet-scale filtering beyond classical Krum.

WHERE IT STOPS

- QKD point-to-point links are unsuitable for frequent V2V broadcasts and suffer attenuation.
- Architecture is theoretical; energy and latency are not measured on automotive hardware.

RESEARCH GAP

Build hardware-in-the-loop validation joining real ECUs, automotive traffic, and accessible NISQ-era quantum resources.

CAPSTONE LINK

Retain federated/behavioral IDS ideas while grounding security in deployable gateway hardware.

PUF • KYBER • DILITHIUM

Lightweight Cryptography + PQC + PUFs for In-Vehicle Communication

WHAT THE PAPER ENABLES

- Combines hardware identity, quantum resistance, lightweight AEAD, and network segmentation.
- Offloading keeps resource-constrained ECUs viable with low communication overhead.

WHERE IT STOPS

- One ProVerif identity query remains unresolved; authentication occurs only at startup.
- No physical automotive benchmark, continuous monitoring, or explicit per-message freshness.

RESEARCH GAP

Authenticated ECUs may become malicious after startup; continuous runtime behavioral detection is missing.

CAPSTONE LINK

Add a lightweight gateway IDS using timing, frequency, spoofing, flooding, and communication patterns.

LSTM • CONTEXT • CAN

Intrusion Detection System for Controller Area Network

WHAT THE PAPER ENABLES

- Context-aware evidence reduces false positives and separates benign anomalies from attacks.
- Timing-aware resistance forces attackers to manipulate several correlated signals.

WHERE IT STOPS

- Limited manufacturers/features; some datasets could not support robust function construction.
- No authentication/cryptography integration; evaluation uses only sensitivity and specificity.

RESEARCH GAP

Evaluate behavioral IDS alongside authenticated, encrypted, and post-quantum-secured CAN gateways.

CAPSTONE LINK

Use cryptography for trust establishment and LSTM monitoring for post-authentication compromise.

IDS for In-Vehicle Networks: Protocols, Applications & Challenges

WHAT THE PAPER ENABLES

- Comprehensive protocol-to-IDS view with future technology directions.
- Connects detection accuracy to practical deployment and migration constraints.

WHERE IT STOPS

- Real-world datasets are limited and heterogeneous.
- Large models and computationally intensive PQC exceed many current ECU capabilities.

RESEARCH GAP

Backward-compatible migration needs lightweight AI, secure gateway inference, PQC authentication, and hardware acceleration.

CAPSTONE LINK

Centralize PQC and IDS at an intelligent gateway while preserving legacy ECUs.